

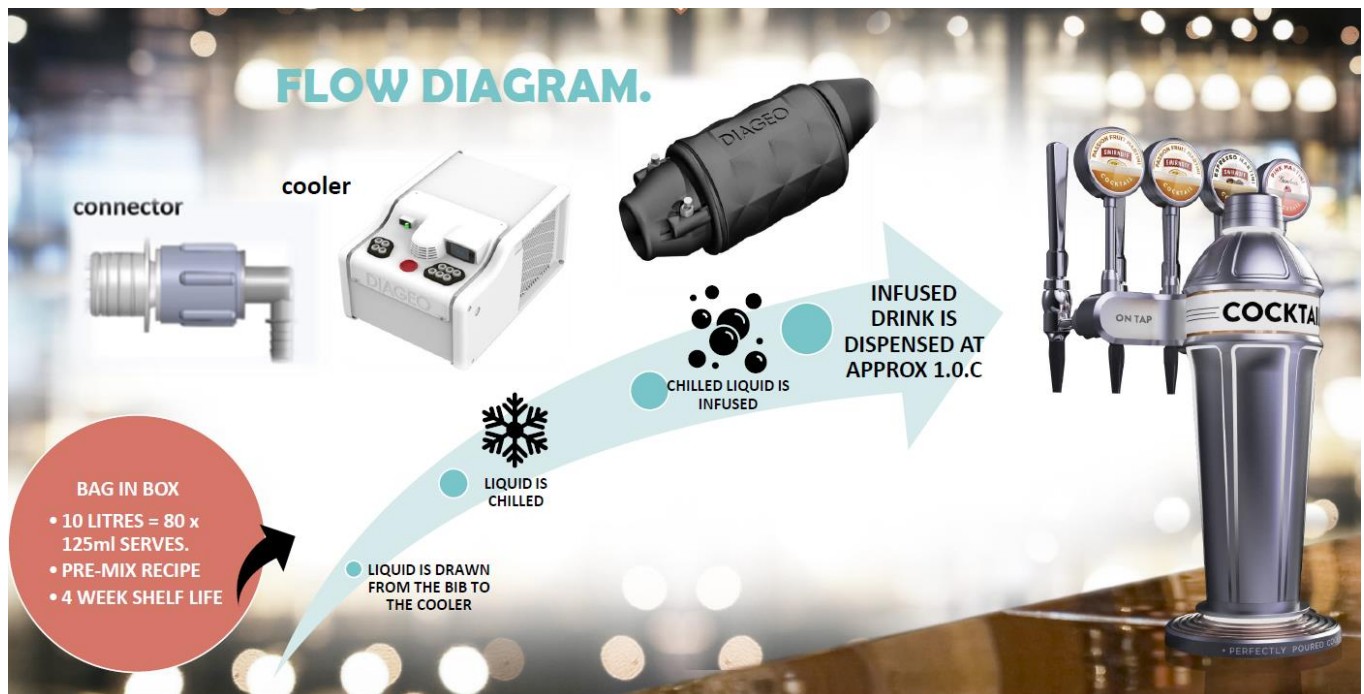
Diageo Europe Partner Markets Draught Cocktail Installation Guide



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System Overview

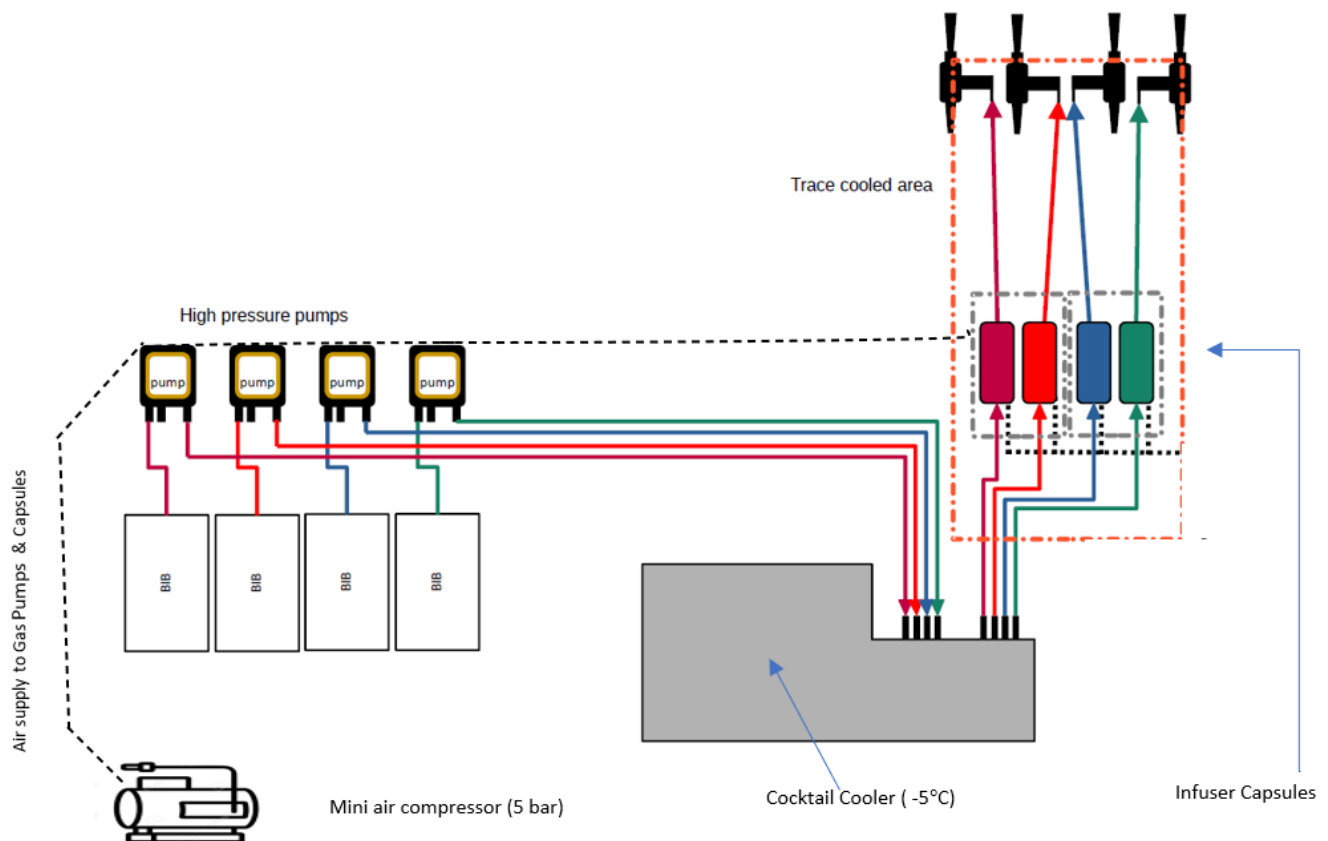


INTRODUCTION TO THE DRAUGHT COCKTAIL DISPENSE SET UP.

HOW IT WORKS.

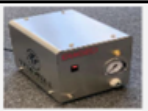
- BAG IN BOX LIQUIDS ARE POSITIONED ON THE SHELF UNDER THE BAR.
- THEY USE A BAG IN BOX CONNECTOR SYSTEM
- LIQUID IS DRAWN FROM THE BIBS AND DELIVERED THROUGH THE SYSTEM BY A T5000 FLOWJET GAS PUMP.
- THE GAS PUMP IS MOUNTED UNDER THE COUNTER AND DRIVEN BY A MINI AIR COMPRESSOR
- THE COMPRESSOR SUPPLIES PRESSURE TO BOTH THE PUMP AND BUBBLEBOX CAPSULE.
- LIQUID ENTERS THE SHELF COOLER (SET AT -5.0) USING A DIGITAL CONTROLLER
- THE COOLED LIQUID THEN ENTERS THE BUBBLEBOX CAPSULE WHERE IT IS INFUSED
- THE BUBBLEBOX CAPSULE HOUSES A CARBONATING DEVICE THAT SERVES 2 TAPS.
- THE INFUSED AND COOLED LIQUID IS POURED THROUGH A FREE FLOW TAP WITH A 25 THOU 5 HOLE DISC SPOUT
- THE LIQUID IS TRACE COOLED TO THE BACK OF THE TAPS.
- THE FONT IS A BESPOKE MANHATTAN 4 TAP FONT USING STANDARD GUINNESS NITRO TAPS.

SYSTEM SCHEMATIC



NB: Each gas pump should have an air inlet shut off valve to allow it to be isolated independently.

SURVEYING FOR A 4-TAP INSTALLATION KEY COMPONENTS & CHECKLIST.

Cooler E400		Bottom Shelf 50mm air gap all around Cooler close to taps <3m W35 x H25.6 x D55 (cms)
Electrical Sockets		3 required: Cooler, Air Compressor, illumination transformer
Mini Air Compressor		Ideally, close to system but can be remote W19 x H19 x D26 (cms)
Manhattan Font		Clamp on tower (max 100mm) < 3m from the cooler Needs to be illuminated Drip tray widest point 32cms
Bag in Box Liquids (BIB's)		Close to cooler but can be separated BIB's Must be Horizontal W20 x H20 x D32 (cms)
Gas Pumps		Site under counter close to BIB's & Cooler 1 pump per tap. Ideally locate behind BiB's W16.5 x H18 x D8 (cms)
Bubblebox Capsule		Site under counter between cooler & font 1 Bubblebox capsule serves 2 taps Can be installed in any orientation

Notes

- Ideally, the cooler should be located under or almost under the taps and < 3m from taps. Consider air circulation around the cooler
- Bag in box liquids, positioned nearby and always horizontal with the Nitro capsule units behind the bag in boxes.
- 3 electrical plug sockets are required:
 - 1st Cooler
 - 2nd Air compressor within 2.5 metre cable length of the cooler
 - 3rd Illumination transformer for the tower
- Air compressor can be sited away from the installation and air pipe work extended to meet Nitro capsule if required.
- There will be two pythons from the cooler, one to each bubblebox capsule.

INSTALLATION.

For ease it is recommended that the inlet section is connected first and the steps follow as listed below.

1. Place the cooler and the font in position
2. Connect capsule inlet to cooler python
3. Connect capsule outlet to fount python.
4. Connect cooler python to cooler.

COOLER

- Model: E400
- Position the cooler below or near to the taps but always on the bottom shelf.
- Once in position then add Glycol:Water at 30:70 mix as instructed below.
- The waterbath capacity is 7.5 litres Note: Cooler weight is 21kg when empty circa. 29kgs when full

NB: Water bath takes approx. 3 hours from room temp to reach -5°C on initial installation

COOLER CONNECTIONS



Connections

1. All connections will be made with JG straight fittings
2. The unit has insulation in place JG connections will butt up to this
3. JG fittings should be insulated with 42mm Armaflex – it is tight but the Armaflex can be compressed to form a tight bundle
4. Product lines and recirc is then combined into single loom.
5. Coolant filler cap once removed gives access to the main coolant bath.

Armaflex in place for tank line insulation
– JG fittings butt onto this (A)

Control Panel



Controller (A)

Electronic controller pre set to -5.0C

Cleaning Switch (B)

Cleaning Switch in the 1 position means normal cooler operation. Cleaning switch in the 0 position means cleaning mode – cooler will not operate.

Setting Cleaning Cycle

Refrigeration/cooling can be switched off via this switch when line cleaning. The switch will be in the 0 position when cleaning in operation

Best to make these connections after the connections to the bubblebox capsule.



insulation

5. Install the product feed pipes ensuring they are colour coded so identifiable back on the outlet of the gas pump and bag in box connector.

6. Install 35 x 19mm Armaflex on the product inlets (left hand side) or use single 5/16th Armaflex to product feed pipework from gaining heat pickup from the cooler.



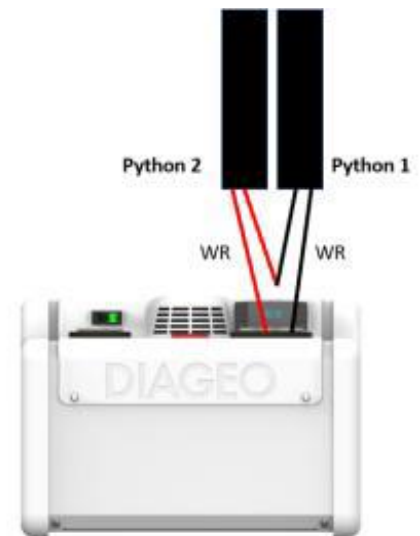
Glycol Recirculation needs to be split to feed both pythons at the cooler as pictured below when installing 3-4 taps.

- Black trace lines go to the top of the tower
- Red trace lines go to the base of the tower.

NB

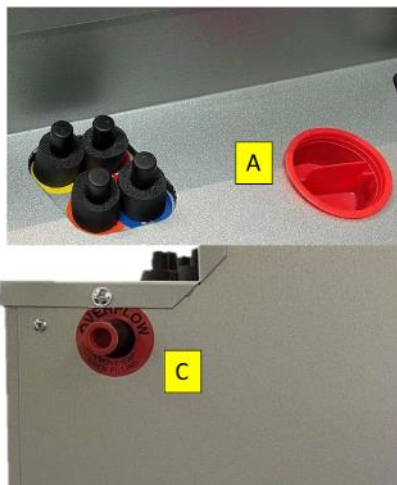
Take care to insulate all pipe work and associated connections and seal the python correctly at every junction to ensure correct temperatures are achieved and no condensation leading to ice formation.

Best practice is to wrap the 2 pythons leaving the cooler together to improve seal and prevent condensation/ice formation.



Filling the cooler

The E400 cooler has a 7.5L water bath and requires a 30% glycol to 70% water mix. -> 2.5 Litres of glycol per cooler.



Filling the D400 cooler

1. Remove filler cap - funnel directly into the tank (A)
2. Remove overflow bung

1. The mix is 30%/70% glycol to water
2. Pour half of the 5litre container into the tank (B)
3. Fill the remainder of the tank with water until the overflow is reached (C)
4. Use a refractometer to check the Glycol mix.
5. Replace bung.



AIR COMPRESSOR.

There are two options

MGF: Mini Bev FPS



Stanwell: Mini Compressor



- The Mini Bev and Stanwell air compressors are oil free clean air mini compressors specifically designed for drinks dispense.
- Conveniently sized for under bar systems and quiet with an operating noise level of 50db meaning any operating noise is lost in normal ambient noise of a pub or restaurant.
- For comparison: Normal conversation is 60db, a refrigerator is 50db, a vacuum cleaner is 70db
- No need for gas bottles or venting of the pump exhaust.
- **The compressor should not be sited on top of cooler.**
- Best placed at the rear of the bottom shelf out of reach.
- Can be located further away if plug sockets require this and air supply tube extended to reach the system.
- The air pressure on the compressor should be set to **5 bar**.
- Lock off knurled nob when set

AIR SUPPLY

- The air is required to feed the Gas pumps and each bubble box unit.
- Air supply can be in series or in parallel using blue 3/8" tubing.
- **John Guest shut off valves should be fitted to each gas pump to allow for isolation if required.**
- Feed air to capsules should be split using 3/8" to 5/16" Y-piece allowing separate feed each capsule.
- No pump exhaust required

MANHATTAN FONT

- The unit is a clamp on tower for up to 100mm thick counter.
- The font comes complete with trace cooling fitted.
- The trace cooling pythons are designed to fit with the Diageo nitro capsules.
- There are 4 x 3/16 colour coded product tubes
- There are 2 x 3/8" colour coded trace cooling looms, black goes to top of the tower red goes to base of the tower.
- Fit the standard Guinness gate taps and elegance handles.
- No compensator valves or Vari-flow taps needed.
- Check 5 hole 25 thou Guinness nozzles are in use.
- Secure lenses and connect font lighting
- Note- all lens pods require a lens. If two products dispensing, use duplicate lenses to create dummy positions.
- Keep branding in pairs.

DIAGEO BUBBLE BOX CAPSULE INTRODUCTION AND INSTALLATION GUIDE.

CONTENTS.

- THE CAPSULE OVERVIEW
- PART NUMBERS
- INSTALLATION OVERVIEW
- PUMP PANEL



THE CAPSULE OVERVIEW

- The capsule is a light weight insulated housing containing 2 carbonators, associated check valves, gas pipework, water recirculation and adjustable needle valves to adjust the level of infusion.
- The capsule is designed to be used with the draught cocktail cooler python (part-904383) and the Manhattan draught cocktail font.
- A single capsule is needed for two products dispensing connected to one of the Manhattan font trace pythons.
- The capsule is to be installed under counter and in-line between the font and the cooler.
- The capsule can be installed at any position or angle. Secure where possible.
- Each capsule comes with the contents below.
- End caps have diameters associated with the fount and cooler python sizes.

BOX CONTENTS.



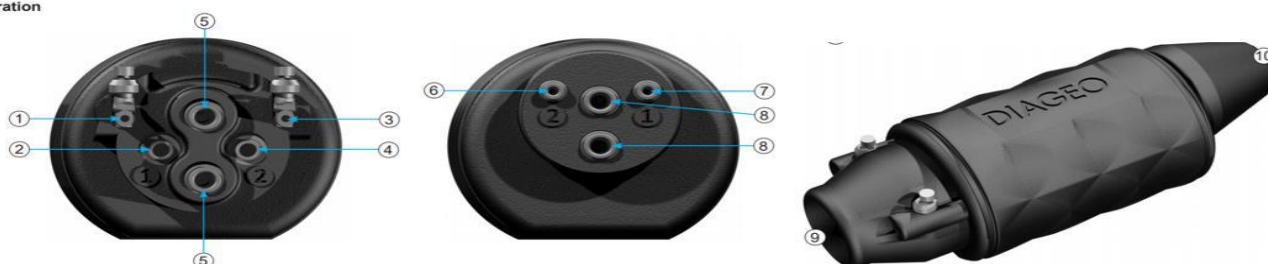
- CAPSULE BODY
- INLET END CAP
- OUTLET END CAP
- 4MM PIPE
- JG STEM CONNECTORS 4MM X 5/16
- RED LOCKING CLIPS
- ROTAREX DOCUMENT

CAPSULE DIMENSIONS.

TOTAL INCLUDING END CAPS FITTED.

L- 337 mm x W- 122 mm x H- 130 mm

Illustration



Legend

① Needle valve 1 Ø 4 mm	⑥ Beverage outlet connection 2 Ø 3/16"
② Beverage inlet connection Ø 8 mm	⑦ Beverage outlet connection 1 Ø 3/16"
③ Needle valve 2 Ø 4 mm	⑧ Recirculation connections Ø 3/8"
④ Beverage inlet connection Ø 8 mm	⑨ Inlet cap
⑤ Recirculation connections Ø 3/8"	⑩ Outlet cap

INSTALLATION.

For ease it is recommended that the inlet section is connected first and the steps follow as listed below.

1. Connect capsule inlet to cooler python
2. Connect capsule outlet to fount python.
3. Connect cooler python to cooler.

STEP ONE- CAPSULE INLET CONNECTION.

- With the cooler python routed but not secured connect the capsule inlet as pictured below.
- Slide the end cap onto the cooler python.
- Slice the cooler python just a few inches to allow the water and product tubes to fan out.
- Connect the water recirc and product tubes. **Make a note of tube colours to inlet number**
- Ensure clean cuts and tubing is pushed fully into place.
- Ensure as much insulation as possible is maintained within the end cap to avoid separation at the joint but allow the cap to fit back to the capsule.



PRODUCT TUBES CONNECTED
WR TUBES CONNECTED
RED LOCKING RINGS FITTED.



There are:

- 2 x 5/16" product line connections
- 2 x 3/8" trace cooling connections
- 2 x 4mm air line connections
- All connections should be secured using the red lock clips



IMPORTANT.
AIM TO MAINTAIN THE
PYTHON DIAMETER.

THE PYTHON DIAMETER
HELPS INSULATE THE END
OF THE CAPSULE AND
PREVENTS AIRFLOW TO
COLD PIPEWORK.

AVOID TIGHT WRAPPING
WITH BLACK TAPE.

TEST END CAP IN POSITION.
NOTE OPEN INSULATION ON PYTHON.
RE-SEAL PYTHON INSULATION AVOIDING
CRUSHING OF THE INSULATION.
MAKE SURE ENOUGH INSULATION IS KEPT
INSIDE THE END CAP TO AVOID CAPS.



SLIDE END CAP BACK DOWN TO
WRAP PYTHON



INSERT THE 4MM GAS TUBING THROUGH THE END CAP AND INTO THE NEEDLE VALVES.



SLIDE HOME THE END CAP. POP INTO POSITION.



FIT THE 4MM STEM ENLARGERS TO THE BLACK PIPE.

STEP TWO- FOUNT PYTHON.

Repeat the process for the capsule outlet connecting to the fount python.

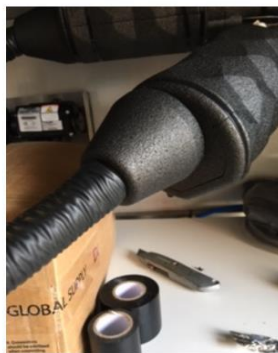
This step is much easier due to the free movement of the trace cooling insulation that can be pulled back rather than cut.



SLIDE ON OUTLET END CAP TO FOUNT PYTHON. PULL BACK PYTHON INSULATION.



CONNECT WR TUBES AND 3/16 PRODUCT TUBES. MAKE A NOTE OF OUTLET NUMBER FOR ASSOCIATED PRODUCT.



SLIDE BACK INSULATION UP TO CAPSULE. SLIDE HOME END CAP. POP INTO POSITION.

**DO NOT
EXTEND
FOUNT
PYTHONS.**

THIS WILL
SLOW FLOW AT
THE TAP.

CAPSULES BEST
POSITIONED
CLOSE TO
FOUNT AS
POSSIBLE.

When the capsule is fully finished the needle valves are still fully accessible

STEP THREE- CONNECT THE COOLER.

- Connect the product inlets using 5/16" straights (E400 cooler)
- Connect the cooler python. Product lines with 5/16" straights recirculation lines with 3/8" straights.
- Pay attention to the insulation requirements of the E400 cooler connections to avoid condensation/ice and maintain best cooling contact.
- Ensure all speed fits are insulated as best practice.
- Follow the guidance for connecting the capsule to cooler python to the cooler previously shown ensuring python integrity.

COMPRESSOR AIR SUPPLY

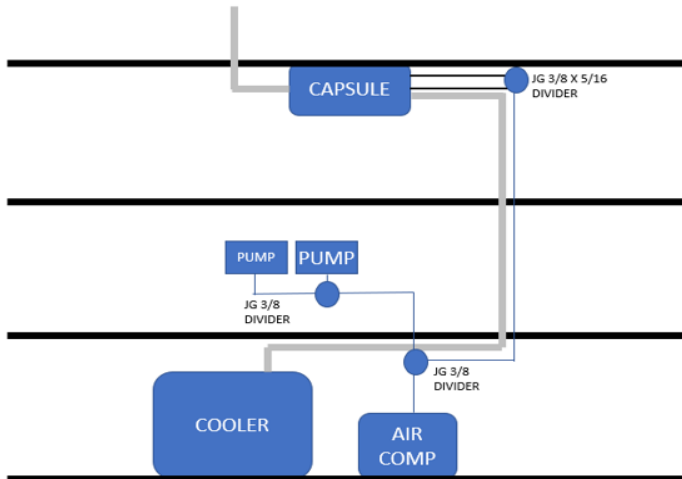
Compressor pipework is configured as below.

Air supply to the pumps and capsules can be in series or in parallel.

For a 4-tap installation repeat for second capsule.

Do not connect air supply to unused pumps or capsules. Alternately isolate these pumps and capsules using the JG shut off valves.

Only pumps and capsules that are in use should have active air supply.



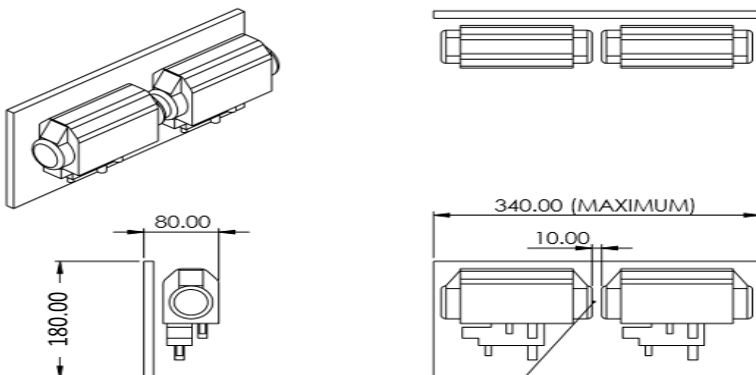
RUN PIPE WORK FROM COMPRESSOR. FOLLOW PYTHON ROUTE TO CAPSULE. USE 3/8 X 5/16 DIVIDER TO CONNECT CAPSULE GAS INLETS.

SPLIT SUPPLY TO FEED GAS PUMP PANEL.

SPLIT TO FEED INDIVIDUAL GAS PUMPS.

GAS PUMPS.

- The T5000 bag in box pump should be secured to the counter or a board secured to the counter.
- A good location for the gas pumps is usually behind the bag in boxes
- Note- maximum distance from the bag to the pump would be around 3 metres.
- Gas pump dimensions H 18 x L 17 x D 8 (cm's)
- The unit clips on and off its backing plate
- Product line inlet secures using a jubilee clip product line outlet secures using a John Guest 3/8" to 5/16"
- Air inlet connects using a 3/8" to 3/82 John Guest
- **John Guest shut off valves need to be fitted to the air supply for individual gas pumps.**



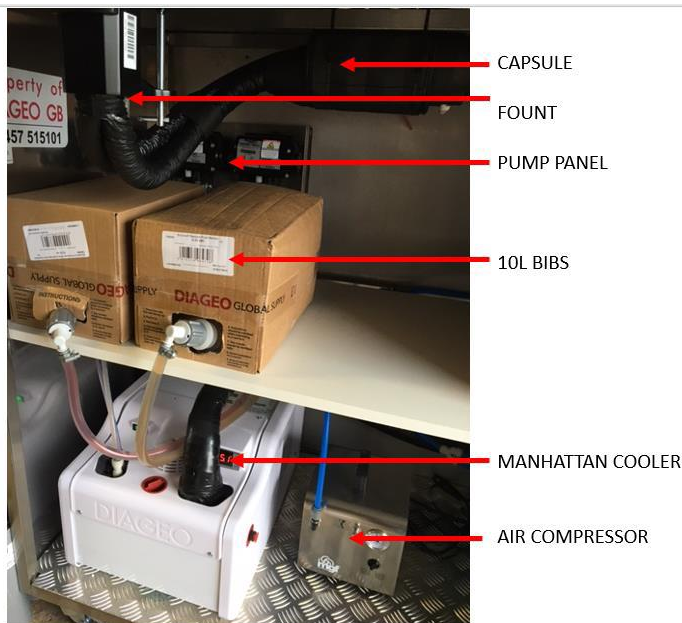
PRODUCT TUBING.

- Connect the PVC hose to the BIB connector using a jubilee clip or crimp.
- Cut to size required allowing for length to reach into bucket when cleaning.
- Connect this into the gas pump inlet and secure with a jubilee clip.
- Secure hose to avoid strain or pulling on pump inlet.
- Air tubing from the pump to the bubblebox will be in place.
- Using 5/16 flavour lock tubing connect the product line from the gas pump outlet to the cooler inlet.

COOLER.

- Connect the product inlets using 5/16 straights (E400 cooler)
- Connect the cooler python.
- Pay attention to the insulation requirements of the E400 cooler connections to avoid condensation and maintain best cooling contact.
- Ensure all speed fits are insulated as best practice.

INSTALLATION LAYOUT EXAMPLE:

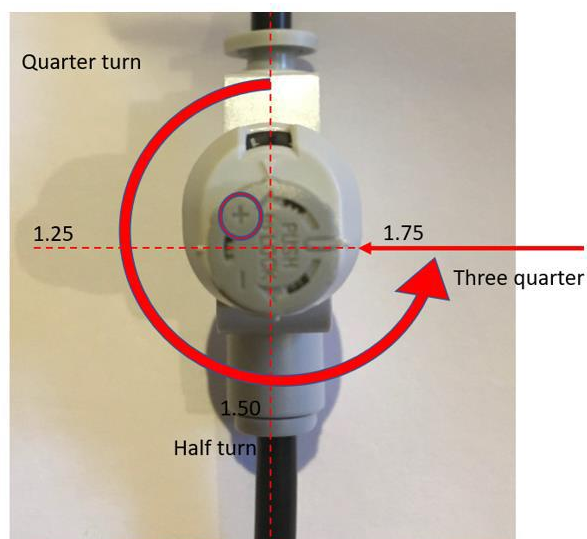
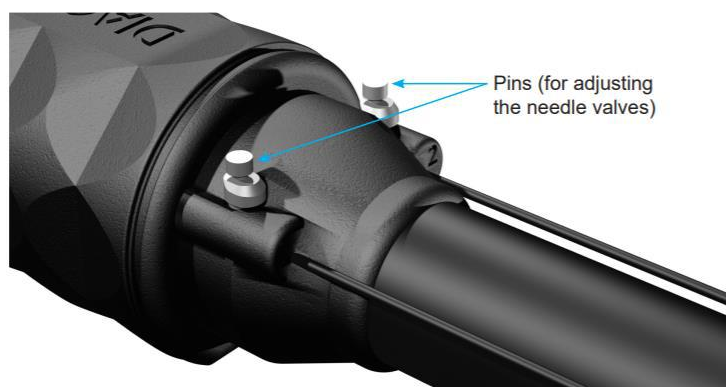


COMMISSIONING/TESTING.

- Cooler filled with 30% glycol 70% water.
- Cooler set to -5.0.c with a 0.5.c differential.

COMMISSIONING/ TESTING.

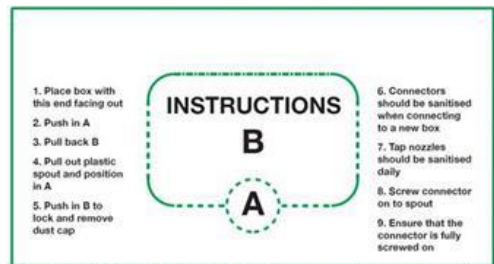
- Needle valves are now fully accessible without opening the capsule.
- **Needle valve setting required at 1.50.** Please check this on installation. Needle valves are not pre-set. Some adjustment may be required further to achieve desired foam spec.
- Foam can be increased/decreased to meet the 7mm – 10mm drink spec by increasing the value on the needle valve if required.
- Lift needle valve dial. Turn clockwise or anti clockwise as required and push down on the dial once set to lock.
- **CHECK ALL PIPEWORK FROM THE ECOMPRESSOR FOR LEAKS.**



CONNECTION AND FINAL CHECKS

- **Bubblebox needle valve set between 1.25 and 1.75)**
- Compressor set to 5 bar- lock off knurled nob when set
- Connect bib for one liquid flavour only.
- Open air shut off valve for gas pump.
- Open tap and pull through liquid without nozzle.
- Add nozzle and confirm settle and head formation.
- Target dispense temperature is 1°C
- Repeat for product 2

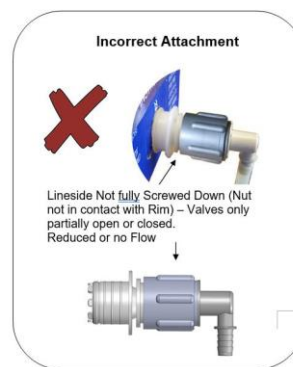
BIB CONNECTOR



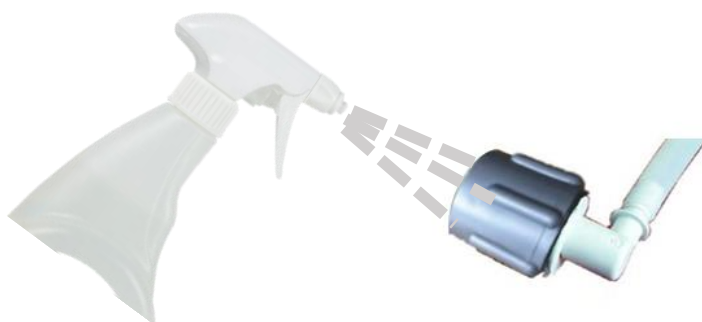
Open the BIB and extract the BIB threaded connector

Step 1
Remove Dust Cap

Step 2
Screw On Lineside
Ensure the lineside is fully screwed down



- A sanitizing spray should be used on the BIB connectors between bag in box changeovers as best practice to maintain the cleanliness of the seal on the connector.
- To use, simply spray lightly onto the connector as pictured below. Connect and continue use.



NOZZLE HYGIENE

At the end of each day nozzles should be removed and cleaned with fresh water and stored dry or put back on the taps. At least once per week they should be removed and soaked in a sanitising solution to help maintain the cleanliness of the internal parts and removal of residue.

All nozzles in the bar will benefit from routine removal and cleaning with a sanitising tablet and clean water to help prevent the spread of infection into the dispense system.

FINISHED DRINK SPECIFICATIONS.

SERVE SIZE

- 125ml is recommended serve size
- The Diageo coupe glass allows 125ml of liquid to 10mm of crema/foam
- 125ml in glass will give 80 drinks per box.

HEAD HEIGHT

- For both passionfruit and espresso, we aim for a 7mm- 10mm head of foam.
- Note: It is recommended to fill the glass a few mm below the rim to allow for head formation and increase when garnish is added.

TEMPERATURE

- The temperature in glass should be 1° C +/-1° at average throughput and with the cooler at -5.0c.
- Some sites may pre-chill glassware. This is very beneficial to final drink temperature

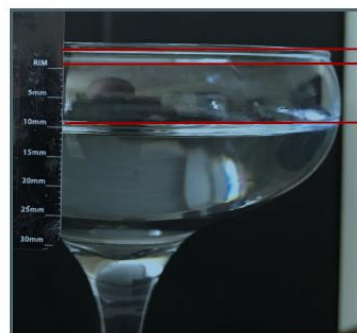
POUR

- The dispense system when correctly set should deliver a nitro style serve into the glass.
- Passionfruit is a foamier looking flow of liquid but still settles to form the correct head.
- Espresso will pour and look similar to Guinness.
- The desired head formation is between 7mm – 10mm.
- Passionfruit forms a good cushion to support garnish, espresso has a smooth surface texture.
- Increasing the head size above 10mm depth can result in undesirable larger bubbles on the surface of the foam more visible in the espresso due to the colour of the head.

GLASSWARE

- Coupe style cocktail glass.
- The Diageo coupe glass presents the best-looking espresso and passionfruit/pink gin martini cocktail.
- The curvature of the glass allows for best head formation and retention.

Serve measurements indicated below.



SPACE BELOW RIM
UP TO 10MM OF FOAM

COCKTAIL LIQUID

Cocktail Descriptions

Passionfruit Martini: 10% ABV

Description:

- Smirnoff Vodka combined with juicy passion fruit and citrusy lime flavours, is balanced with smooth notes of vanilla and marshmallow

Garnish

- A slice of passion fruit

Gordon's Pink Martini: 10% ABV

Description:

- A fruity blend of juicy raspberry and tart lemon, with red berries and a sweet taste of Gordon's Pink

Garnish

- 1 Raspberry

Espresso Martini: 12.5% ABV

Description:

- A rich and smooth coffee-flavoured cocktail with characteristic vanilla and chocolate notes, made with Smirnoff Vodka

Garnish

- 3 Coffee beans.

Captain Morgan Strawberry Daiquiri: 10% ABV

Description:

- This smooth cocktail mixes Captain Morgan's Spiced Caribbean Rum with sweet ripe strawberries, juicy lime, and rich notes of vanilla and rum spice.

Garnish

-

Some examples of finished serve



DRAUGHT COCKTAIL DISPENSE- CLEANING AND HYGIENE.

- Recommended line cleaning frequency for your draught cocktail dispense system is every **2-4 weeks**.
- This line clean should ensure the refresh of dispense system components whilst maintaining a low loss of liquid to the retailer which is also aided by the short run of the under-bar system.

BEFORE GETTING STARTED- ensure the cleaning of the system is being completed by a competent and trained member of staff.

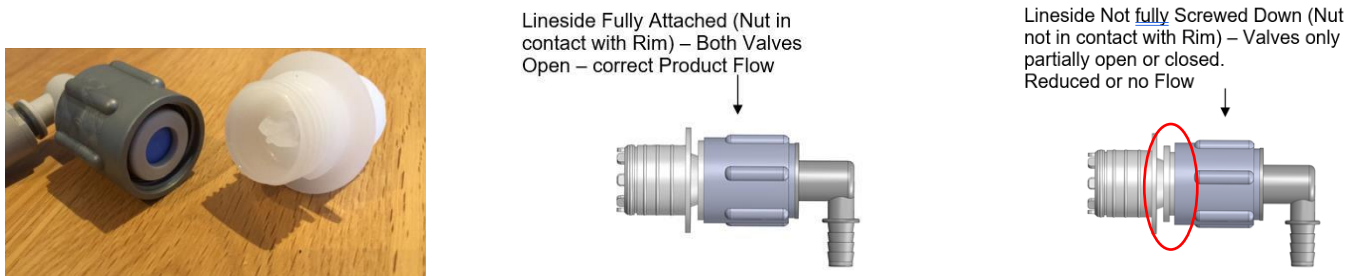
RECOMMENDED PPE SHOULD BE USED DURING THE CLEANING OF THE DRAUGHT BEER DISPENSE SYSTEM.

IMPORTANT

- Due to the sub-zero operating temperatures of the glycol/water bath it is critical that the cooler is switched off at least 2 hours before flushing the system with water or ideally, the night before line cleaning is due.
- Ensure water has been replaced with cocktail product when turning cooler back on to avoid freezing
- **Never interrupt the air supply to the capsules at the compressor or needle valve when detergent is in the system.**

CLEANING PROCEDURE

- Remove the BIB connector and fit the BIB cleaning adaptor ensuring it's screwed fully home.
- Note this adaptor is the same tap/connection supplied with the bag in box, so spares can be obtained by simply removing a tap from a finished empty bag if ever required.



WHEN PROPERLY CONNECTED THIS OPENS THE TAP SO LIQUID CAN PASS THROUGH THE SYSTEM.

1. Place the connector into a bucket or container of fresh water and flush the system removing all cocktail liquid.
2. Similar in practice to the cleaning process of beer.
3. When the system is flushed with clean water, replace the water with the correct diluted measure of beer line cleaner and repeat the flush of the system.
4. Follow guidelines for recommended soak times and dilution as instructed on the cleaning detergent manufacturer label.
5. After the recommended soak time period, flush the system continuously with fresh clean water.
6. Recommended 5 litres per tap to ensure cleaning detergent is flushed.
7. Line clean is now complete.
8. Reconnect the product to the top bag in box connector and pull up product to the tap.
9. Now, turn the cooler back on.
10. Cooler temperature will be displayed, this will now reduce to -5.0.c ready for serving drinks at optimum temperature.
11. **Important Note:** If the cooler is turned on when water is in the system this will lead to freezing and blocked lines and delay in serving drinks whilst natural thawing takes place.

If there is an interruption to the air supply to the capsules when detergent is in the capsule then system should be flushed and capsule should then be replaced and returned to the workshop for flushing out.

COOLER and STANWELL COMPRESSOR SERVICE.

Cooler Grill Cleaning

The fan cover is designed to be removed to allow for easy periodic cleaning



Draught Cocktail Compressor 5 minute service.

Check and clean the aspiration filter inside the cover following the instructions below:

Remove R/H side casing from the compressor as shown below using your 3mm Hex key.

Unscrew



Lift

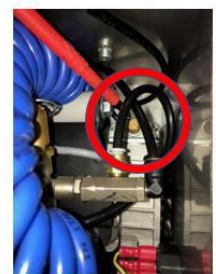
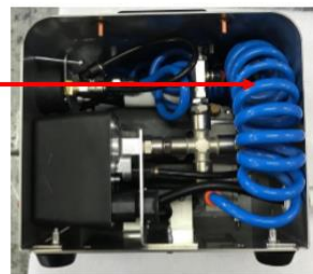
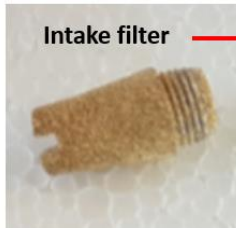


Remove



- Locate and remove intake filter. (unscrew)
- Blow through with airline / clean with water.
- Then reinstate.

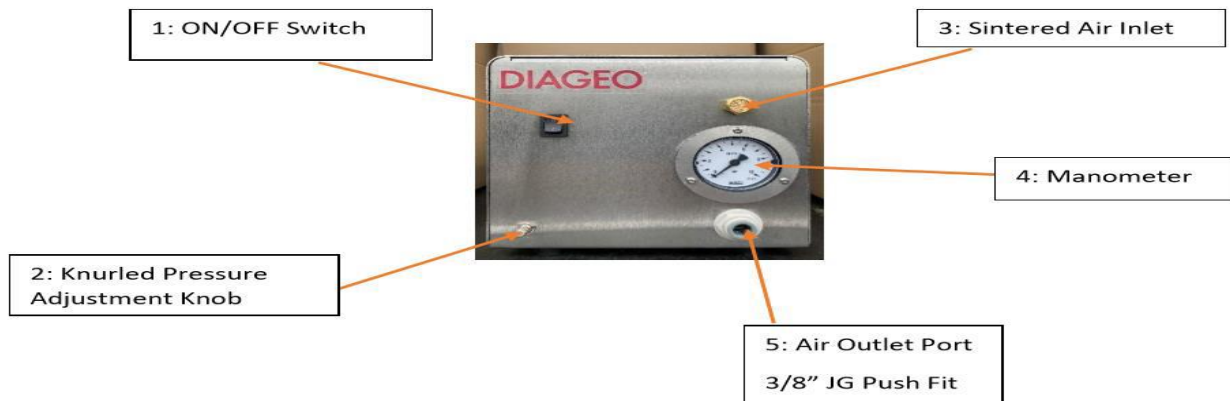
Intake filter



Appendix 1

Air Compressor & Gas pump information

Overview of Controls



Maintenance

- The sintered filter design prevents any large particles in the air from being pulled into the system. Any large particles will fall off the filter during times when the system is quiescent.
- As a result, the filter should be cleaned using a small, DRY brush annually
- A five-year full clean is recommended where the filter is removed and blown through with an air-line. To do this will require the compressor casing to be removed, allowing access to the back of the filter. Tools required would be a 15mm spanner for the outside of the filter and a 17mm spanner to hold the fitting inside the casing.

Rear View of the Stanwell Mini Compressor



Compressor

T5000 BAG IN BOX PUMP.

The t5000 pump has a vacuum switch which are referred to as the “auto-shut off”. The auto shut-off is a separate block incorporating the product inlet port and the gas inlet port.

The auto shut-off on the T5000 series is the part that sits under the main pump body.



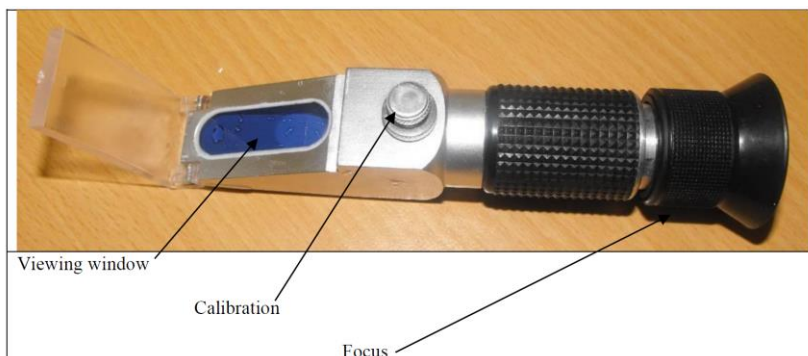
HOW IT WORKS

- As the bag empties, the pump suction causes a vacuum in the product inlet line. As soon as the vacuum is sensed, the vacuum switch operates and shuts off the gas flow to the pump, so the pump stops. Connecting a new bag releases the vacuum and gas is allowed to flow again to drive the pump.
- The main reason for the vacuum switch is to stop the pump from running continuously and using all the gas. If there was no vacuum switch, the line to the dispense valve would eventually have no product in it, so the pump would not stall even with tap closed (because there would be no build-up of static pressure beyond the pump).
- Under normal circumstances, the static pressure in the line beyond the pump (when the dispense valve is closed) basically stalls the pump when that static line pressure equals the drive pressure into the pump from the gas bottle or compressor. As the valve opens, the static pressure drops to become dynamic pressure and the pump is free to run, but if there is no product in the line, this will not happen as the lack of liquid in the line will not build any pressure, hence the reason for the vacuum switch in bag in box.
- As there is no air in the bag, no air gets into the line as the vacuum is created in the bag as soon as it is empty.
- Note:
 - Maximum distance from the bag to the pump would be around 3 metres.
 - Shut off valves should be fitted to each pump air inlet to allow for isolation when no BIB liquid is attached.

APPENDIX 2

REFRACTOMETER

CAN BE USED ON INSTALLATION AND SERVICE TO CHECK 30/70 GLYCOL MIX

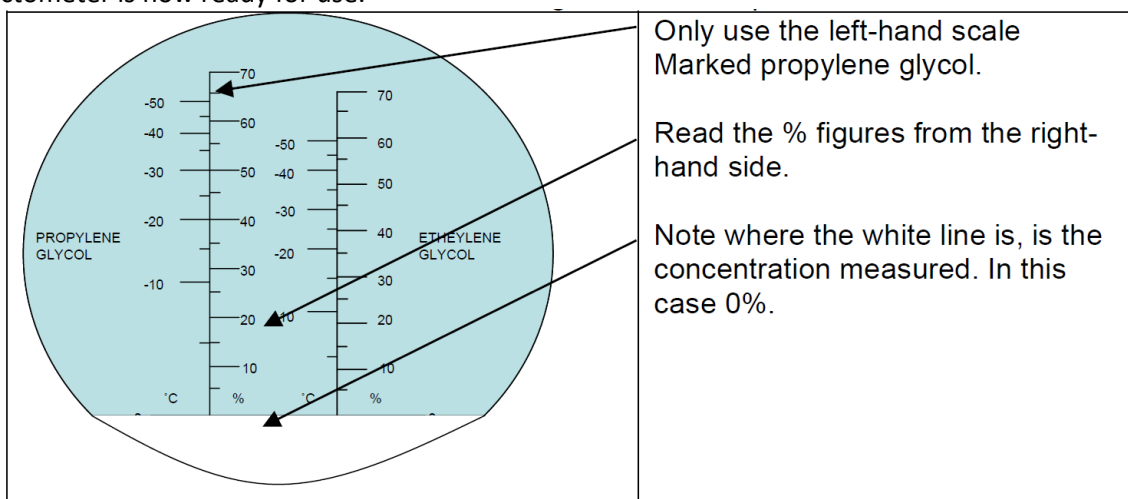


The refractometer is used to measure the refraction of light through a liquid. The version above has a scale included to measure the concentration of Propylene glycol in the liquid. It can be used to measure the concentration of glycol in a Draught Cocktail cooler. It can also be used to measure the concentration of glycol in the hot side of a water split remote cooler. Calibration is essential before using.

Calibration

Before using the refractometer, the calibration should be checked:

1. Clean the viewing window to ensure there is no liquid on it.
2. Apply a few drops of water on the viewing window and close the plastic cover to keep them in place.
3. Looking through the eye piece hold the refractometer up to the light.
4. You should see something similar to the picture below
5. Using the calibration screw adjust the level until it is showing 0%.
6. You can then lock the screw to prevent movement.
7. The Refractometer is now ready for use.



Measuring

Measuring is identical to calibration but always ensure that the viewing window is clean and free from liquid. Take a sample from the cooler bath or glycol tank and place on the viewing area.

Hold the refractometer up to the light and read off where the background colour turns from blue to white on the left-hand scale.